Chapter 9

1. Core Definitions

$$\boxed{\mathbf{p} = m\mathbf{v}} \qquad K = \tfrac{1}{2}mv^2 = \frac{p^2}{2m} \Rightarrow m = \frac{p^2}{2K}$$

Momentum is a vector in the same direction as velocity.

2. Impulse–Momentum Theorem

$$\mathbf{F}_{\text{net}} = \frac{d\mathbf{p}}{dt} \qquad \mathbf{J} = \int_{t_i}^{t_f} \mathbf{F}\,dt = \Delta\mathbf{p} \qquad \mathbf{F}_{\text{avg}} = \frac{\Delta\mathbf{p}}{\Delta t}$$

Area under F–t curve gives impulse:

$$J_x = \int F_x(t)\,dt = \Delta p_x$$

$\Delta p = m(v_f - v_i)$

Impulse approximation (short collisions):

$$\mathbf{F}_{\text{avg}} \approx \frac{\Delta\mathbf{p}}{\Delta t}$$

3. Constant Acceleration (impact / stop)

$$v_f = v_i + at \qquad v_f^2 = v_i^2 + 2a\Delta x \qquad t = \frac{v_f - v_i}{a} \qquad F_{\text{avg}} = m\frac{\Delta v}{\Delta t}$$

4. Momentum Conservation

$$\sum \mathbf{p}_i = \sum \mathbf{p}_f$$

$$\sum \mathbf{F}_{\text{ext}} = \frac{d\mathbf{P}_{\text{sys}}}{dt} \quad \Rightarrow \quad \text{If } \sum \mathbf{F}_{\text{ext}} = 0,\ \mathbf{P}_{\text{sys}} = \text{constant}$$

Angle of motion:

$$\tan\theta = \frac{v_{fy}}{v_{fx}}$$

5. Vector & System Relations

$$p_{\text{net}} = \sqrt{p_x^2 + p_y^2} \qquad \tan\theta = \frac{p_y}{p_x}$$

$$\mathbf{P}_{\text{tot}} = M\mathbf{V}_{\text{cm}} \qquad \mathbf{V}_{\text{cm}} = \frac{\sum m_i\mathbf{v}_i}{\sum m_i}$$

$$\sum \mathbf{J}_{\text{ext}} = \Delta\mathbf{P}_{\text{sys}}$$

6. Kinetic Energy of a System

$$K_{\text{tot}} = \sum \tfrac{1}{2}m_i v_i^2$$

Even if $\mathbf{P}_{\text{tot}} = 0$, total kinetic energy can be nonzero.

7. Elastic Collision (1D)

$$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2}v_{1i} + \frac{2m_2}{m_1 + m_2}v_{2i}$$

$$v_{2f} = \frac{2m_1}{m_1 + m_2}v_{1i} + \frac{m_2 - m_1}{m_1 + m_2}v_{2i}$$

Relative velocity reverses:

$$v_{1f} - v_{2f} = -(v_{1i} - v_{2i})$$

8. Totally Inelastic (stick together)

$p = mv \Rightarrow v = \frac{p}{m}$: momentum speed

$v^2 = 2gh \Rightarrow h = \frac{v^2}{2g}$

$J_x = \frac{1}{2}(\text{base})(\text{height})$ $\qquad \Delta p_x = J_x$

$$v_f = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2} \qquad \Delta K = -\frac{m_1 m_2}{2(m_1 + m_2)}(v_{1i} - v_{2i})^2$$

9. General Inelastic

Momentum conserved, kinetic energy not.

$$\Delta K = K_f - K_i$$

10. Recoil / Exchange (bullet, boxer, etc.)

$$m_A v_{A,i} + m_B v_{B,i} = m_A v_{A,f} + m_B v_{B,f} \qquad \mathbf{F}_{\text{avg}} = \frac{m(\mathbf{v}_f - \mathbf{v}_i)}{\Delta t}$$

11. Oblique Bounce (angle change)

$$|\Delta\mathbf{p}| = 2mv_i\cos\theta$$

12. Center of Mass (coordinates)

$$\mathbf{R}_{\text{cm}} = \frac{1}{M}\sum_i m_i\mathbf{r}_i \qquad M = \sum_i m_i$$

$$X_{\text{cm}} = \frac{1}{M}\sum m_i x_i \qquad Y_{\text{cm}} = \frac{1}{M}\sum m_i y_i$$

13. Rocket Motion

Differential form:

$$m\,dv = -v_{\text{ex}}\,dm$$

Integrated (Tsiolkovsky):

$$\Delta v = v_{\text{ex}}\ln\left(\frac{M_i}{M_f}\right)$$

Thrust and power:

$$F_{\text{thrust}} = -\dot{m}v_{\text{ex}} \qquad P = F_{\text{thrust}}\,v = \dot{m}\,v_{\text{ex}}\,v$$

14. Quick Solvers / Ratios

$$\frac{p_1}{p_2} = \frac{m_1 v_1}{m_2 v_2} \qquad m = \frac{p^2}{2K} \qquad v = \frac{2K}{p}$$

Rain mass in Δt:

$$m = \rho(\text{depth})(\text{area})\frac{\Delta t}{T}$$

Ship stopping:

$$a = \frac{v_f^2 - v_i^2}{2\Delta x} \qquad F_{\text{avg}} = m\frac{\Delta v}{\Delta t}$$

15. Signs & Directions

Choose a positive axis. Bounce reversal:

$$\Delta p = m(v_f - v_i)$$

A negative result indicates reversal of direction.

16. Units & Constants

$$1\,\text{g} = 10^{-3}\,\text{kg} \qquad 1\,\text{mg} = 10^{-6}\,\text{kg} \qquad 1\,\text{cm} = 10^{-2}\,\text{m}$$

$$\rho_{\text{water}} \approx 1000\,\text{kg/m}^3$$

$a = \frac{v_f^2 - v_i^2}{2\Delta x}$ $\qquad F = ma$ $\qquad J = F\Delta t$

$V_f = \frac{J}{M}$

m = mass

Chapter 10

1. Angular Position / Velocity / Accel

$\theta = s/r$, 1 rev $= 2\pi$ rad $= 360°$. $\omega = \frac{d\theta}{dt}$, $\alpha = \frac{d\omega}{dt}$; $\omega_{\text{avg}} = \Delta\theta/\Delta t$, $\alpha_{\text{avg}} = \Delta\omega/\Delta t$. ω[rad/s] $= 2\pi f$, $\omega = \frac{2\pi}{60}$rpm, $\theta[°] = \theta[\text{rad}]\,\frac{180}{\pi}$.

2. Constant-α Kinematics

$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$, $\omega = \omega_0 + \alpha t$, $\omega^2 = \omega_0^2 + 2\alpha\,\Delta\theta$.

3. Linear–Angular Relations (r)

$v_t = r\omega$, $a_t = r\alpha$, $a_r = v^2/r = r\omega^2$. Period: $\omega = \frac{2\pi}{T} = 2\pi f$.

4. Torque, Work, Power

$\tau = rF_\perp = rF\sin\phi$, $\sum\tau = I\alpha$. $W = \int \tau d\theta (= \tau\Delta\theta)$, $P = \tau\omega$. $\tau_{\text{net}} = \sum r_i F_i \sin\phi_i$ (CCW +).

5. Angular Momentum & Impulse

$L = I\omega$, $\sum\tau = \frac{dL}{dt}$. Particle: $L = rp\sin\phi$. If $\sum\tau_{\text{ext}} = 0 \Rightarrow I_i\omega_i = I_f\omega_f$.

6. Moment of Inertia (MoI)

$I = \sum m_i r_i^2 = \int r^2 dm$. Parallel axis $I_P = I_{\text{cm}} + Md^2$; Perpendicular $I_z = I_x + I_y$. Hoop MR^2; Disk $\frac{1}{2}MR^2$; Annulus $\frac{1}{2}M(R_1^2 + R_2^2)$; Sphere $\frac{2}{5}MR^2$; Rod(center) $\frac{1}{12}ML^2$; Rod(end) $\frac{1}{3}ML^2$.

7. Rotational Kinetic Energy & Work–Energy

$K_{\text{rot}} = \frac{1}{2}I\omega^2$, $dK = \frac{1}{2}\omega^2 r^2 dm$. $W_{\text{net}} = \Delta K_{\text{rot}}$.

8. Rolling Without Slipping

$v_{\text{cm}} = R\omega$, $a_{\text{cm}} = R\alpha$. $K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2$. $v_{\text{top}} = 2v_{\text{cm}}$, $v_{\text{bot}} = 0$. $mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$, $v = R\omega$.

9. Pulleys / Drums / Mixed Radii

No slip $a_t = r\alpha$. $\sum\tau = \sum(r_j T_j) = I\alpha$. Relate T, a, α via FBD of masses.

10. Friction Braking / Grindstone

$\tau_f = \mu_k N r$, $\alpha = \tau_{\text{net}}/I$. $\omega_f = \omega_i + \alpha t$, $\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$, turns $= \Delta\theta/(2\pi)$.

11. Polynomial Angle Models

If $\theta(t) = a + bt + ct^2$: $\omega = b + 2ct$, $\alpha = 2c$. Area under $\omega(t)$=$\Delta\theta$; revs =$\Delta\theta/2\pi$.

12. Linear ↔ Rotational Analogy

Linear → Rotational: $s \to \theta$, $v \to \omega$, $a \to \alpha$, $F = ma \to \tau = I\alpha$, $p = mv \to L = I\omega$, $W = Fd \to W = \tau\theta$, $K = \frac{1}{2}mv^2 \to K_{\text{rot}} = \frac{1}{2}I\omega^2$.

13. RPM / Conversions & Quick Links

ω[rad/s] $= 2\pi$(rev/s), rpm $= 60$(rev/s), $\omega = \frac{2\pi}{60}$rpm. Revs: revs $= \frac{1}{2\pi}\int \omega\,dt$.

14. Problem Shortcuts (WebAssign Set)

1 of 2

up/down $\alpha = \tau/I$ then kinematics. Energy $K_{\text{rot}} = \frac{1}{2}I\omega^2$. Parallel/perpendicular axis for offset rotations. Two-radii pulley $r_{\text{in}}T_{\text{in}} - r_{\text{out}}T_{\text{out}} = I\alpha$, $a = r\alpha$. Conserve L: $I_i\omega_i = I_f\omega_f$ (skater, star).

15. Key / Symbol Legend

θ = angular displacement (rad)
ω = angular velocity (rad/s)
α = angular acceleration (rad/s²)
t = time (s)
r, R = radius (m)
s = arc length (m)
v_t, v_{cm} = tangential / linear speed (m/s)
a_t, a_r, a_{cm} = tangential / radial / linear acceleration (m/s²)
F = force (N)
τ = torque (N·m)
I = moment of inertia (kg·m²)
M, m = mass (kg)
K, K_{rot} = kinetic energy (J)
L = angular momentum (kg·m²/s)
μ_k = kinetic friction coefficient
T = tension (N)
P = power (W)
W = work (J)
h = height (m)
Subscripts $_{i,f}$ = initial/final
Subscript $_{\text{cm}}$ = center of mass

$\omega = \frac{2\pi}{T}$ $\qquad \omega = \frac{\Delta\theta}{\Delta t}$

$\theta = \frac{s}{r}$

$\omega = 2\pi f$

f = rotational frequency

Chapter 11

1. Angle–Arc–Speed Links

$\theta = s/r$. $\quad \omega = v/r$, $\alpha = a_t/r$.

Units: ω[rad/s] $= 2\pi f = \frac{2\pi}{60}$ rpm.

2. Rolling (no slip) & Energy

$v_{\text{cm}} = R\omega$, $a_{\text{cm}} = R\alpha$.

$K = \frac{1}{2}Mv^2 + \frac{1}{2}I_{\text{cm}}\omega^2 = \frac{1}{2}Mv^2\left(1 + \frac{I_{\text{cm}}}{MR^2}\right)$.

Common $1 + I/(MR^2)$: hoop 2, solid cyl. 3/2, solid sphere 7/5.

Incline (no slip): $a_{\text{cm}} = \dfrac{g\sin\theta}{1 + I/(MR^2)}$.

3. Rolling with Slipping (μ_k)

Transl.: $Ma = Mg\sin\theta - f_k$, $\quad f_k = \mu_k Mg\cos\theta$.

Rot.: $I\alpha = f_k R$. $\quad$ If $v \neq \omega R$: $\alpha = (f_k R)/I$.

Solid cylinder: $I = \frac{1}{2}MR^2 \Rightarrow \alpha = \dfrac{2\mu_k g\cos\theta}{R}$.

4. Yo–Yo / Unwinding (string radius r)

$Mg - T = Ma$, $\quad Tr = I\alpha$, $\quad a = r\alpha$. $\quad \Rightarrow a = \dfrac{Mg}{M + I/r^2}$.

Solid cyl. $I = \frac{1}{2}Mr^2 \Rightarrow a = \frac{2}{3}g$ (down).

5. Particle Angular Momentum & Torque

$\mathbf{L} = \mathbf{r} \times \mathbf{p} = m\,\mathbf{r} \times \mathbf{v}$, $\quad L = mr_\perp v$.

$\tau = \mathbf{r} \times \mathbf{F} = \frac{d\mathbf{L}}{dt}$.

System: $\mathbf{L}_{\text{tot}} = \sum m_i(\mathbf{r}_i \times \mathbf{v}_i)$, $\quad \tau_{\text{ext}} = \frac{d\mathbf{L}_{\text{tot}}}{dt}$.

6. Rigid Body Angular Momentum

About symmetry axis: $L = I\omega$. $\quad$ About point P: $L_P = I_P\omega$, with $I_P = I_{\text{cm}} + Md^2$.

Pure rolling about contact C: $L_C = (I_{\text{cm}} + MR^2)\omega$.

7. Conservation of Angular Momentum

If $\sum\tau_{\text{ext}} = 0$, then $L_i = L_f$.

Coupled rotors/disks: $I_1\omega_1 + I_2\omega_2 = (I_1 + I_2)\omega_f$ (signs by direction).

Inelastic hit at rim (mass m sticks at R): $\quad mvR = (I_{\text{body}} + mR^2)\omega_f$.

8. KE Changes in Rotational Processes

$K = \frac{1}{2}I\omega^2$. $\quad$ Coupling/sticking: energy lost to internal work/friction.

Stopping rolling object: $W_{\text{stop}} = K_{\text{total}}$.

Bug crawling radially: L const., K changes (static friction does work).

9. Quick Vector Shortcuts

$\mathbf{r} = (x, y, z)$, $\mathbf{v} = (v_x, v_y, v_z) \Rightarrow \mathbf{L} = m\,\mathbf{r} \times \mathbf{v}$.

$\mathbf{r} \times \mathbf{v} = (yv_z - zv_y)\hat{\imath} + (zv_x - xv_z)\hat{\jmath} + (xv_y - yv_x)\hat{k}$.

$\tau = \mathbf{r} \times \mathbf{F}$ (same determinant). $\quad +\hat{k}$ = out of page.

10. Gyroscope / Precession

$L = I\omega$. $\quad \tau = Mgr_{\text{cm}}$ (lever arm $\perp$ to g).

$$\Omega_p = \frac{\tau}{L} = \frac{Mgr_{\text{cm}}}{I\omega}.$$

11. Problem Shortcuts (mapped to Ch. 11 set)

Bike/auto wheels: $\omega = v/r$, $\theta = s/r$; $\alpha = a_t/r$.

Marble/cylinder (no slip): $a = g\sin\theta/(1 + I/(MR^2))$.

Incline w/ slip: $a = g\sin\theta - \mu_k g\cos\theta$, $\alpha = (\mu_k Mg\cos\theta R)/I$.

Rolling up ramp: energy with $v = R\omega$ for top speed/height.

Work to stop: $W = \frac{1}{2}Mv^2\left(1 + \frac{I}{MR^2}\right)$.

$r-p$, $r-F$: $\mathbf{L} = \mathbf{r} \times (m\mathbf{v})$, $\tau = \mathbf{r} \times \mathbf{F}$.

Four-particle L: sum vectors carefully (signs!).

Coupled rotors: conserve L; then ΔK.

Mass sticks to rim: $mvR \to (I + mR^2)\omega_f$; $K \downarrow$.

Bug ↔ center: $I\omega = \text{const} \Rightarrow \omega \propto 1/I$.

Precession: $\Omega_p = Mgr_{\text{cm}}/(I\omega)$.

1. Core Definitions

$$\mathbf{p} = m\mathbf{v} \qquad K = \tfrac{1}{2}mv^2 = \frac{p^2}{2m}$$

Momentum is a vector in the same direction as velocity.